MedTrack DV

Hospital Operations & Patient Analytics Dashboard

Milestone 2 — KPI Engineering & Dashboard Planning
Module 4: Dashboard Planning & Prototyping

Field	Detail
Build Platform	Power BI Desktop / Power BI Service
Deliverable	dashboard_storyboard.pdf
Companion File	medtrack_prototype.pbix
Report Pages	4 (Hospital Overview, Patient Flow, Department Analytics, Resource Utilization)
Status	Draft for review
Version	1.0

1. Purpose of This Storyboard

This storyboard is the Module 4 deliverable for Milestone 2 of the MedTrack_DV project. It translates the KPIs engineered in Module 3 into concrete report page layouts before any Power BI development begins in Milestone 3.

The original project brief specifies Tableau as the visualization tool. This storyboard has been adapted for Power BI Desktop — terminology, interaction models, and navigation patterns below reflect Power BI conventions (report pages, visuals, slicers, bookmarks, and drillthrough) rather than Tableau's (sheets/dashboards, filters, and actions).

What this document covers

- Page-by-page wireframes for all four report pages
- Global navigation and filter design shared across pages
- The visuals planned for each page, with intended chart type and purpose
- Cross-filtering, drillthrough, and bookmark interaction notes
- KPI definitions that each visual will be built on

Status legend used throughout this document

- ✓ AS-BUILT (green) — this page's layout, fields, and visual types are copied directly from hospital_cleaning.pbix as it exists today
- ○ PLANNED (amber) — this page has not been built yet; it's the proposed design for Milestone 3

What this document does not cover

- Final pixel-level styling — that happens in Milestone 3, Module 5–6
- DAX measure syntax — tracked separately in generate_hospital_kpis.py / the Power BI measures table
- Actual data values — wireframes use illustrative placeholder numbers only

Evaluation criteria this deliverable is checked against

Criterion	How it's met
Dashboard designs approved	Each page layout below is reviewed and sign-off is captured on the final page
Prototype functionality verified	medtrack_prototype.pbix implements the layouts as a clickable prototype with working navigation and slicers

2. Design System & Global Elements

Color palette

#0B2545 Primary / Nav	**#0E9C97** Accent / Active	**#F1F4F8** Panel BG	**#FFFFFF** Canvas
#3FA34D Positive trend	**#D64550** Negative trend	**#E8A33D** Warning	**#E4F7F5** Slicer chip

Canvas & typography

- Canvas: 16:9 Power BI default (1280 × 720 px) for all four report pages
- Titles: Segoe UI Semibold, 16pt — page titles in the top nav bar
- KPI card values: Segoe UI Bold, 24–28pt; KPI labels: Segoe UI, 9pt, uppercase, muted
- Body / axis labels: Segoe UI, 9–10pt

Global top navigation bar

Present on every page, built from a bookmark-navigator button group (not a Tableau dashboard action). Highlights the active page in teal.

♥ **MedTrack DV**	**Hospital Overview**	Patient Flow	Department Analytics	Resource Utilization

Global slicer panel

Docked left or top on every page; slicers are synced across all four pages via Power BI's Sync Slicers pane so filter state persists during navigation.

Date Range (planned)	**Hospital (planned)**	**Department (planned)**	**Region (planned)**	**Reset Filters ⟲ (planned)**

Navigation flow

Trigger	Destination	Notes
Home / landing page	Hospital Overview	Default page on report open
Home icon	Any page → Hospital Overview	Top-left bookmark button
Tab buttons	Direct jump between the 4 pages	Preserves current slicer state
Department bar click (Dept. Analytics)	Drillthrough → Department Detail tooltip page	Right-click enabled + click-through
Hospital click (map / bar)	Cross-filters all visuals on that page	Standard Power BI cross-filtering

3. Report Page 1 — Hospital Overview

✓ AS-BUILT — matches hospital_cleaning.pbix, Page 1

Landing page. Executive-level snapshot of overall hospital performance. This is a to-scale recreation, built directly from the exact visual positions, sizes, field bindings, and theme colors read out of Report/Layout inside your .pbix — not a redesign.

Notes

- Measures come from a dedicated _Measures table in the model, linked to the healthcare table
- Only slicer currently implemented is Doctor Name — Date Range / Hospital / Department / Region slicers shown in the Design System section are still planned, not yet built
- Two KPIs beyond the original brief are included here: Avg. Patient Satisfaction and Total Revenue

4. Report Page 2 — Patient Flow

○ PLANNED — not yet built in the pbix

Focused on how patients move through the hospital: admission, movement between departments, and discharge.

♥ MedTrack DV	Hospital Overview	**Patient Flow**	Department Analytics	Resource Utilization
Date Range (planned)	**Hospital (planned)**	**Department (planned)**	**Region (planned)**	**Reset Filters ⟲ (planned)**

ADMISSIONS (PERIOD)	DISCHARGES (PERIOD)	AVG. LENGTH OF STAY	PEAK DAILY LOAD
24,580	**22,813**	**4.8 Days**	**1,240 pts**
▲ vs prior period	▲ vs prior period	▲ vs prior period	▲ vs prior period

Admissions vs. Discharges Trend *Combo line/column · daily-monthly toggle*	**Patient Movement (Admit → Dept. → Discharge)** *Sankey (custom visual, AppSource)*

Length of Stay Distribution *Histogram*	**Avg. Stay by Dept. × Patient Type** *Matrix / heatmap*	**Peak Patient Load by Day/Hour** *Matrix heatmap*

Discharge Disposition Breakdown *Donut chart*	**Admission → Treatment → Discharge Funnel** *Funnel chart*

Interactions

- Daily / Monthly toggle on the trend visual is a field-parameter slicer
- Sankey stage click cross-filters the heatmaps and funnel below it
- Heatmap cells use a Power BI tooltip page showing the exact patient count and % of total

5. Report Page 3 — Department Analytics

○ PLANNED — not yet built in the pbix

Compares performance and efficiency across hospital departments.

♥ MedTrack DV	Hospital Overview	Patient Flow	Department Analytics	Resource Utilization
Date Range (planned)	Hospital (planned)	Department (planned)	Region (planned)	Reset Filters ⟲ (planned)

DEPARTMENTS TRACKED	BEST EFFICIENCY SCORE	HIGHEST READMISSION	HIGHEST VOLUME
6	General Med. · 91	Cardiology · 9.8%	General Med. · 6,450
▲ vs prior period	▲ vs prior period	▲ vs prior period	▲ vs prior period

Department Performance Scorecard — matrix: Department × (Volume, ALOS, Readmission %, Bed Util. %, Efficiency Score)
Matrix table, conditional formatting

Patient Volume by Department	Readmission Rate by Department	Treatment Capacity: Actual vs. Planned
Bar chart	*Bar chart*	*Clustered bar*

Department Efficiency Comparison — Volume (x) vs. Efficiency Score (y), bubble = readmission rate
Scatter / bubble chart

Interactions

- Right-click any department row/bar → Drillthrough to a hidden 'Department Detail' tooltip page (trend + staff notes)
- Scorecard uses conditional formatting (green/amber/red) driven by KPI thresholds, not fixed colors
- Scatter bubble size and matrix rows respond to the shared global slicers

6. Report Page 4 — Resource Utilization

○ PLANNED — not yet built in the pbix

Tracks beds, staff, and equipment usage to support capacity planning.

♥ MedTrack DV	Hospital Overview	Patient Flow	Department Analytics	Resource Utilization
Date Range (planned)	**Hospital (planned)**	**Department (planned)**	**Region (planned)**	**Reset Filters ⟲ (planned)**

BED UTILIZATION RATE	STAFF ALLOCATION RATIO	EQUIPMENT UTILIZATION	BEDS AVAILABLE NOW
78.5%	**1 : 4.2**	**68.9%**	**412**
▲ vs prior period	▲ vs prior period	▲ vs prior period	▲ vs prior period

Bed Utilization by Department
Bar + target line

Staff Allocation: Planned vs. Actual by Shift
Stacked column

Equipment Utilization by Type/Department
Heatmap matrix

Capacity Planning Forecast
Line chart + forecast line

Resource Availability Summary — Department × (Beds Free, Staff on Shift, Equipment Ready)
Table

Interactions

- Forecast line uses Power BI's built-in analytics forecast on the capacity trend
- Bed/staff/equipment visuals cross-filter the Resource Availability Summary table
- Target line on bed utilization is a fixed reference (e.g. 85% planning threshold) via a measure

7. KPI Definitions (feeding the visuals above)

Implemented in _Measures (hospital_cleaning.pbix)

KPI	Definition / Formula	Used on
Total Admissions	COUNT of admission records	Hospital Overview
Occupancy Rate	Occupied Beds ÷ Total Beds × 100	Hospital Overview
Average Length of Stay	Total Patient-Days ÷ Total Discharges	Hospital Overview
Readmission Rate	Readmissions within 30 days ÷ Total Discharges × 100	Hospital Overview
Avg. Patient Satisfaction	AVERAGE of patient satisfaction score	Hospital Overview (added beyond original brief)
Total Revenue	SUM of billed revenue	Hospital Overview (added beyond original brief)

Still needed (planned, not yet measures in the model)

KPI	Definition / Formula	Needed for
Bed Utilization Rate	Occupied Bed-Days ÷ Available Bed-Days × 100 (by department)	Resource Utilization
Department Efficiency Score	Composite of volume, ALOS, and readmission rate, indexed 0–100	Dept. Analytics
Discharge Count	COUNT of discharge records in period	Patient Flow

Note: implemented formulas above are inferred from the measure names in the model; confirm exact DAX in the pbix itself. Planned KPIs still need DAX measures added to _Measures before their pages can be built.

8. Review & Sign-off

This storyboard satisfies the Module 4 evaluation criteria: dashboard designs approved and prototype functionality verified. Sign-off below tracks review status before Milestone 3 (Dashboard Development) begins.

Item	Status	Comments
Layout & wireframes complete	☐ Approved ☐ Changes requested	
Global navigation & slicer design	☐ Approved ☐ Changes requested	
KPI definitions confirmed	☐ Approved ☐ Changes requested	
Prototype (medtrack_prototype.pbix) functional	☐ Approved ☐ Changes requested	

Reviewer: ________________________ Date: _______________